1) Smart online Examination Registration system:

a) Input Conditions & Equivalence Classes:

Condition	Valid Input	Invalid Input	Expected output
Course Registration	Registered	Not Registered	Reject
Attendance	75-100%	Below 75% or outside 0-100%	Reject
Fee	Paid	Unpaid	Request Payment
Register Number	Exactly 12 digits	Invalid length or character	Display Error.

B) ECP and BVA

Attendance	Expected Result
74%	Reject
75%	Eligible
76%	Eligible
101%	Invalid Input

ECP test representative values from each class. BVA test values near boundaries, reducing redundant tests while detecting boundary errors.

C) Decision table :-

Course	Attendance ≥ 75%	Fee paid	Valid Register Number	Result
yes	yes	yes	yes	Accept
NO	yes	yes	yes	Reject
yes	NO	yes	yes	Reject
yes	yes	NO	yes	Request Payment
yes.	yes.	yes.	NO.	Display Error.

D) Random Testing :-

Attendance	Register Number	Expected Result
60%	123456789012	Reject
90%	123456789012	Accept if other condition pass
101%	123456789012	Invalid attendance
85%	12345AB89012	Invalid Register Number.

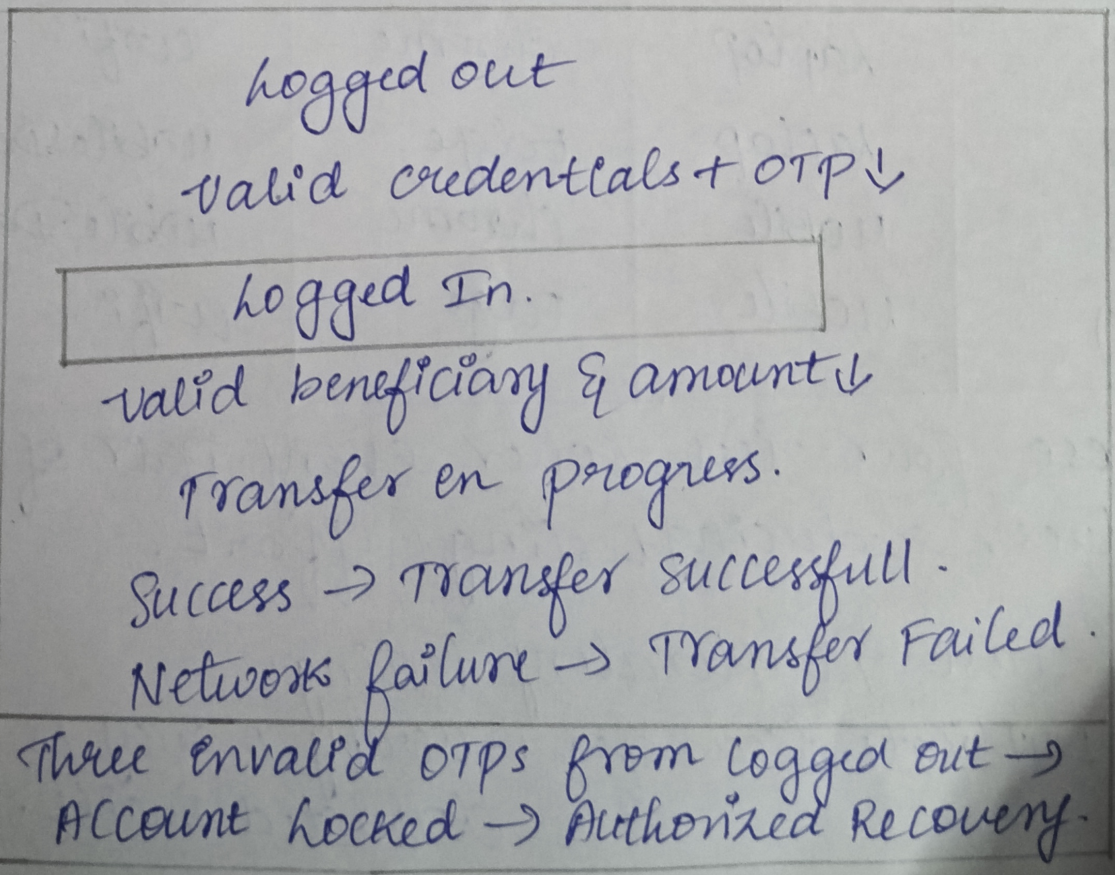
Random enputs can reveal unexpected errors missed by systematic tests.

e) Risk-based testing:-

Prioritize eligibility, payment & register number validation. Test boundary values & combinations of envalid enputs. Evaluate coverage using requirements, equivalence classes, decision-table rules and boundary cases. Fix defects and repeat tests before the release.

2.) Online Banking Application:-

a) State transition Diagram:



Invalid transitions include transferring while logged out and accessing a locked account with recovery.

b) OTP Boundary Testing:

OTP Attempts	Expected Result
Correct on first attempt	login successfull
Two wrong, then correct	Login Successful
Three wrong attempts	Account Locked.

c) Pairwise Testing:

Test	Device	Browser	Network
1	Laptop	Chrome	WiFi
2	Laptop	Edge	Mobile Data
3	Mobile	Chrome	Mobile Data
4	Mobile	Edge.	WiFi.

These four test cover every pair of factor values, reducing testing effort.

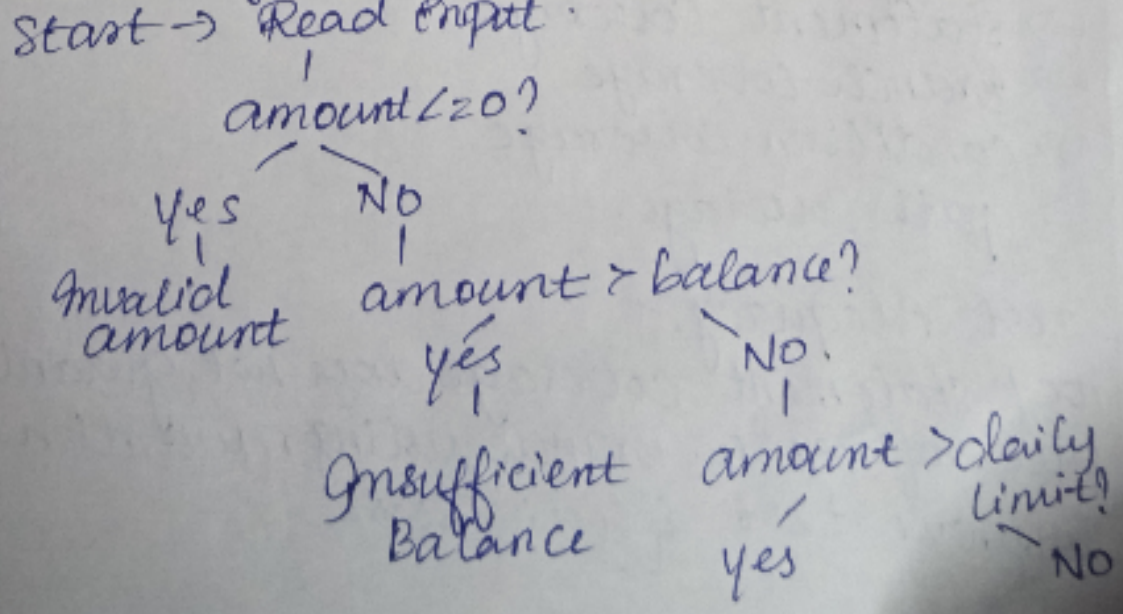
D) Use-Case & Exploratory testing:

→ Duplicate transfer :- verify money is not deducted twice.

→ Session Expiry: Require Login again
→ Network failure: Verify transaction status and safe retry.
→ Wrong beneficiary: Confirm recipient details before transfer.

e) COTS gateway testing :-
Test gateway integration, API responses, browser compatibility, timeout recovery, duplicate requests, encryption, authentication Since source codes unavailable, use Sandbox testing, logs and vendor documentation. Internal defects and cannot be fully inspected.

3) Smart ATM fund-Transfer system:
a) Control flow graph:



Limit error Transfer success.

Stop

B) cyclomatic complexity:

There are 9 nodes, 11 edges & 1 connected component

$V(G) = E - N + 2P = 11 - 9 + 2(1) = 4$

There are 3 decision nodes, so $V(G) = 3 + 1 = 4$

C) Independant paths & Test Cases:

Assume balance = ₹1000 and daily limit = ₹500

Amount	Expected Result
₹0	Invalid Amount
₹1,200	Insufficient Balance
₹700	Daily limit exceeds
₹400	Transfer successful, balance ₹600

D) Coverage techniques:-

→ Statement Coverage.
→ Branch coverage
→ condition coverage
→ path coverage.

E) Test Adequacy:-

100% statement coverage does not guarantee defect free software. Branch testing, mutation testing improve test effectiveness.

4) online login security system:

A) loop and decision structures:

The program uses a while loop for three password attempts and an IF-ELSE decision to check the password.

3) loop testing:-

Test	Input	Expected Result
Zero iterations	count = 3 at loop entry	No password attempt
one iteration.	correct first attempt	login success
Two iteration	wrong, correct	login success
Three iteration	wrong, wrong, correct	login success
Maximum attempts	Three wrong passwords.	Account locked.

) Defect & correction:-

The Reported defect is displaying both success and lock messages.

) coverage Techniques:-

→ condition coverage
→ Branch coverage
→ path testing

e) Mutation Testing:

Mutation	Test to detect
Change < 3 to <= 3	Check for fourth attempt
Remove counter increment	Test repeated wrong password.
Change == to !=	Test correct & incorrect password.

5) Student Result processing system:

a) control flow graph and dataflow:

→ The program uses two nested loops. The outer loop processes students

→ N, M, i, j, mark, total.

b) loop Testing:

Test Input	output
N=0	No total printed
N=101	Reject if mark is 100
M=11	Reject if mark is 10.

C) Data Flow Testing:

Data flow testing checks variable enitiali-zation and usage. N & M control loops, i, j count iteration

D) ECP, BVA, Error Guessing

Input	Valid range	Invalid values
Student	1-100	-1, 101
Subject	1-10	-1, 11
Mark	0-100	-1, 101

E) Test adequecy:

→ Statement coverage
→ Branch coverage
→ loop coverage.